



Parul University
Faculty of Engineering and Technology
Parul Institute of Engineering and Technology

Department: AI-ML/AI-RO/AI/AI-DS/CSE/MICRO/SAP/QUICK/ORACLE/IT/AERO

Subject Name	PROBABILITY, STATISTICS AND NUMERICAL METHOD	A.Y	2025/2026
Subject Code	303191251	Semester	4th

Assignment-4

Sr No	Question	COs	B.T Level	Competence
1	Two dice are thrown at the same time. Find the probability that: (a) the sum is a prime number (b) at least one die shows a multiple of 3.	6	3	Apply
2	A bag contains 5 red, 4 blue and 3 green balls. Two balls are drawn at random without replacement. Find the probability that: (a) both are of same color (b) one is red and one is blue.	6	3	Apply
3	A discrete random variable X has the following distribution: X: 0,1,2,3 and P(X): 0.1, 0.3, 0.4, 0.2. Find: (a) Mean (b) Variance (c) Standard deviation.	6	3	Apply
4	The following data represents marks of 40 students: Class: 0–10, 10–20, 20–30, 30–40, 40–50; Frequency: 3,7,12,10,8. Find: (a) Mean by step deviation method (b) Median (c) Mode.	6	3	Apply
5	Find Karl Pearson's coefficient of correlation for: X: 10,12,15,18,20; Y: 25,28,30,34,36. Comment on the nature of correlation.	6	4	Analyse
6	From the data: X: 2,4,6,8,10; Y: 5,9,11,15,18. Find both regression equations and estimate Y when X = 7.	6	4	Analyse
7	Using Newton's Forward Interpolation Formula, find y at x = 2.5 from: x: 2,3,4,5; y: 8,27,64,125.	6	3	Apply
8	Using Newton's forward difference formula, find dy/dx at x = 1 from: x: 0,1,2,3; y: 1,2,4,8.	6	3	Apply
9	Using Simpson's 1/3 rule, evaluate $\int_0^6 \frac{1}{(1+x^2)} dx$ by dividing the interval into 6 equal parts.	6	4	Analyse
10	Evaluate $\int_1^2 \sqrt{(x^2+1)} dx$ using the trapezoidal rule with 5 equal subintervals.	6	3	Apply
11	Using Euler's method, find y at x = 0.4, taking h = 0.1: dy/dx = x + y, y(0) = 1.	6	3	Apply
12	Using fourth-order Runge-Kutta method, find y at x = 0.2, h = 0.1: dy/dx = x ² + y, y(0) = 1.	6	4	Analyse